Aiden Chen

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EDUCATION

University of Michigan-Ann Arbor **Ann Arbor, MI**
*B.S. in **Computer Science and Statistics*** Exp. May 2028

- **Relevant Coursework:** Discrete Math, Programming & Intro to Data Structures (C++)
- **Extracurriculars:** Michigan Data Consulting (MDC), Iota Omega Epsilon Professional Co-Ed Fraternity

EXPERIENCE

Data Engineering Consultant **Oct 2026 - Present**
American Economic Liberties Project (via MDC) *Ann Arbor, MI*

- Designing a relational schema and backend to replace AELP's static Big Tech Abuse Tracker (200+ entries, 5 companies), scaling to 25+ corporations with 6 filter dimensions (incl. 8 harm categories, 5 jurisdictions) and 3 sort orders.
- Building Python and SQL ETL pipelines that normalize news articles into categorized company records joined with CourtListener and SEC EDGAR API data, plus intake for a public antitrust-reporting questionnaire, so staff can keep it current.

Co-Founder & CTO **Dec 2024 - Nov 2025**
UnitNode *Mullica Hill, NJ*

- Built a Python ETL pipeline that parses unstructured utility emails into structured tenant bills with due dates, loading them into PostgreSQL (Supabase) and feeding Stripe API payment processing.
- Designed the relational data model and workflows for properties, leases, bills, and payments, plus translated tenant alerts.
- Ran user interviews with property management firms and homeowners and shipped weekly iterations from their feedback, scaling the platform to 1,900+ properties and 40+ owners across multiple states within 8 months.

Research Scholar, Data-Flow Analysis & Benchmarking **Jun 2025 - Oct 2025**
NJ Governor's School of Engineering & Technology (Rutgers-College of Engineering) *New Brunswick, NJ*

- Built 32 of 118 apps and their vulnerability-detection plugins in JungleGym, a CWE-labeled benchmark dataset used to evaluate TaintMonkey, a Python dynamic taint analysis library for Flask; published in IEEE Xplore and presented at MIT URTC 2025.
- Added JSON-input tainting and fixed major data-flow tracking bugs in TaintMonkey, which instruments Flask endpoints via monkey patching (no source changes) and fuzzes inputs to catch tainted data reaching unsanitized sinks.
- Wrote the paper's Experimental Procedures, designed all evaluation figures, and built the 13-slide deck presented at MIT URTC. Selected as 1 of 64 statewide scholars (<1% acceptance).

Data Engineering & Automation Intern **Jun 2024 - Sep 2024**
Home Vista Realty *Philadelphia, PA*

- Built a 1,160+ property database across NJ and PA and an automated Python and OpenAI API pipeline parsing Philadelphia Water Department billing reports into structured month-by-month usage records per property.
- Identified the water-cost threshold at which tenants began missing payments and flagged properties approaching it using historical billing data, targeting 12 high-risk Philadelphia homes for low-waste utility upgrades with local contractors.

PROJECTS

Data Analyst | Python (scikit-learn, Pandas, NumPy), SQL, Tableau **Jul 2026 - Aug 2026**
E-Commerce Churn Analysis & Pricing Strategy: ML Prediction & CLV Optimization *Under NDA until W27*

- Trial project for a national gym franchise using a public 18,000-record e-commerce churn dataset as a stand-in for member data; benchmarked Logistic Regression against a Random Forest (0.922 ROC-AUC), flagging 76.2% of churners at 94.2% precision.
- Identified support calls and cart abandonment as top churn drivers and modeled 5/10/20% promotion scenarios across $1M+ at-risk CLV; recommended a targeted 10% offer protecting $50K (∼50% ROI), earning a paid W27 engagement.

Financial Analytics Consultant | Excel, SQL, Tableau **Jun 2026 - Jul 2026**
M&A SME Financial Analysis: 10-Year Baseline & 5-Year Projections *Mullica Hill, NJ*

- Modeled 10 years of financials for a $40.8M-revenue SME and forecast 5 years of Bull/Base/Bear scenarios in SQL and Tableau.
- Advised a pivot from top-line growth to cost discipline: a 3-year margin-improvement phase before pursuing an exit.

ADDITIONAL

Data Engineering & Analytics: SQL (PostgreSQL), Python (Pandas, NumPy, scikit-learn), ETL Pipelines, Data Modeling, Tableau
Business Analysis: Excel, Requirements Gathering, User Interviews, Financial Modeling, Scenario & ROI Analysis
Engineering: TypeScript, C++, Git, REST APIs, Supabase, Vercel, Claude Code, Cursor
Interests: Triathlons, Distance Running (2x Ultramarathons, 2x Marathons), MMA (Jiu-Jitsu World Medalist), Skydiving, Hiking